香港科技大學
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ELEC 3200: System Modeling, Analysis and Control

Lecture 12-14: Stability

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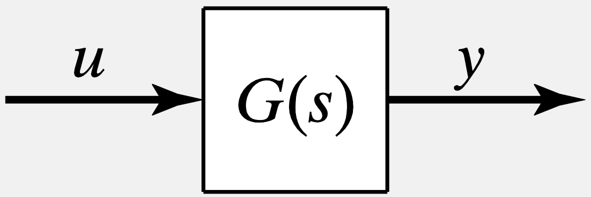
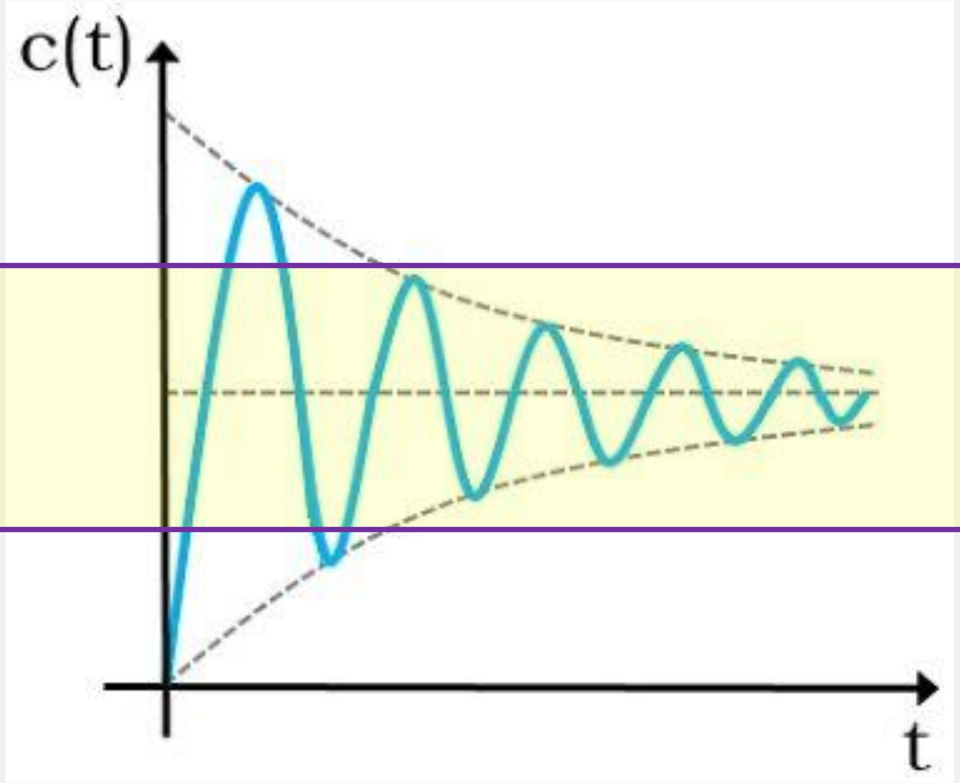
Systems descriptions

- Consider the control system, where $u(t)$ is the input, $y(t)$ is the output, and $G(s)$ is the rational transfer function.

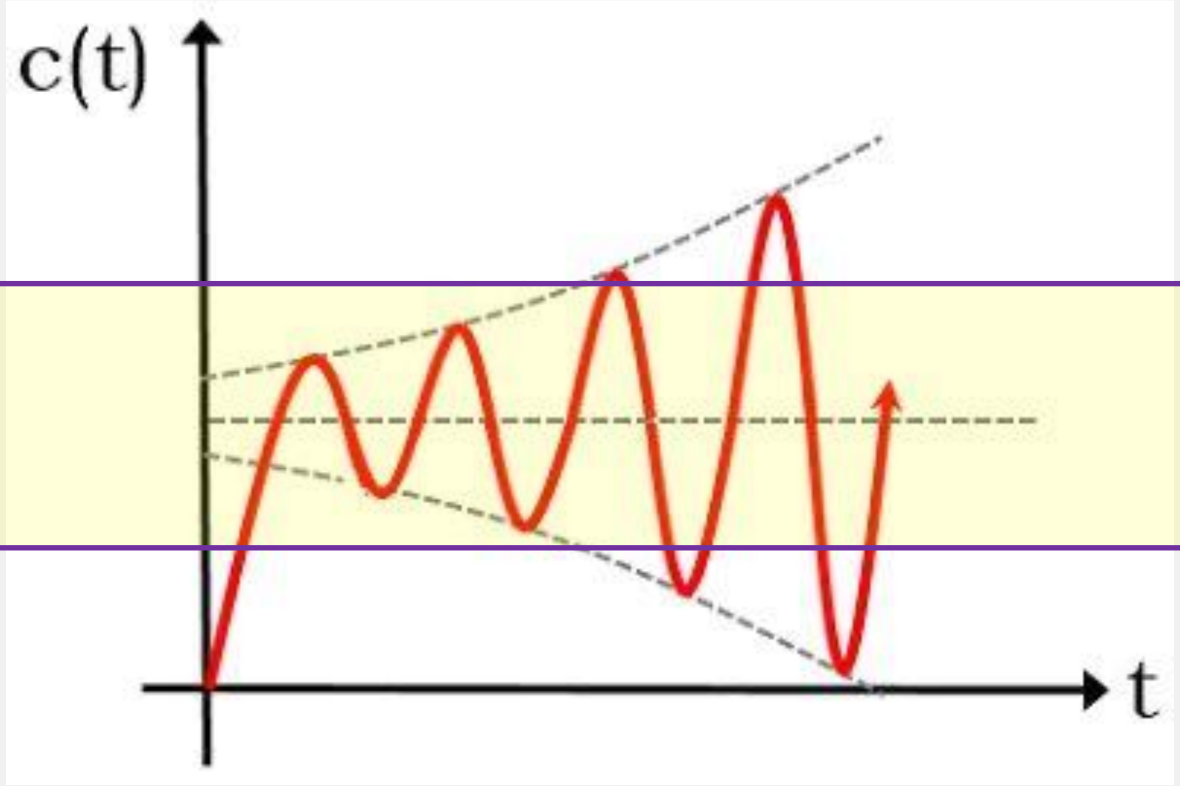


※ For convenience, we simply use $G(s)$ to represent the system which has the transfer function $G(s)$.

Bounded v.s. Unbounded



Bounded



Unbounded

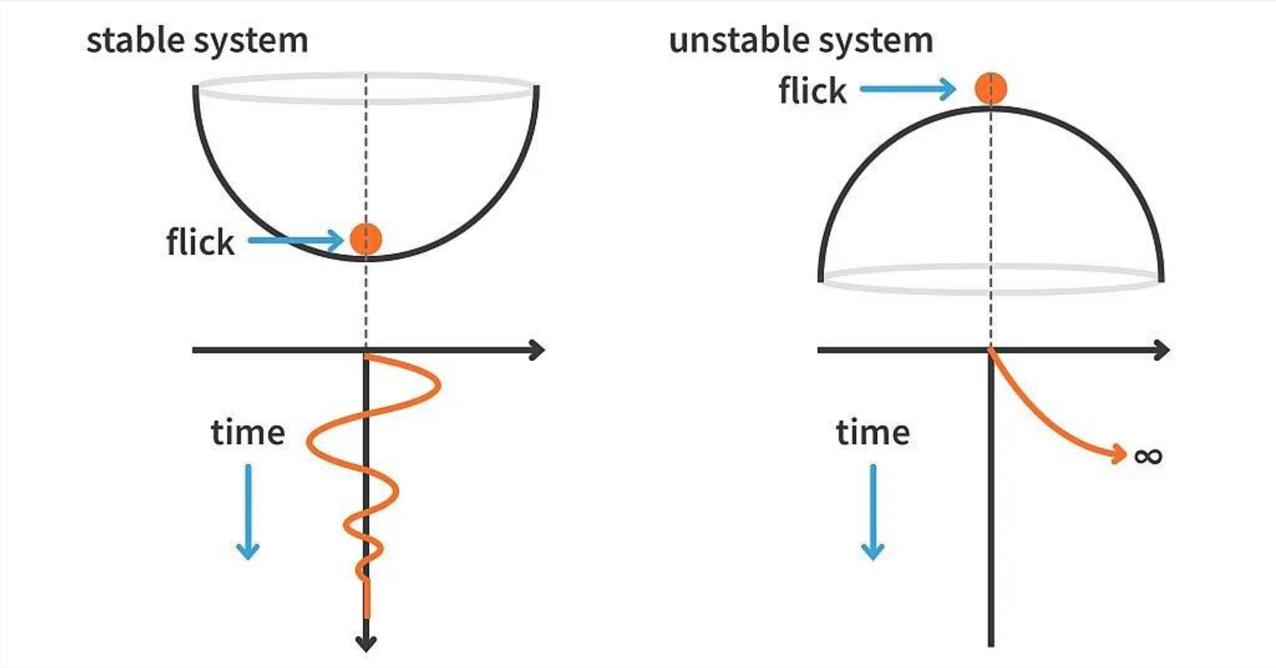
Bounded Signals

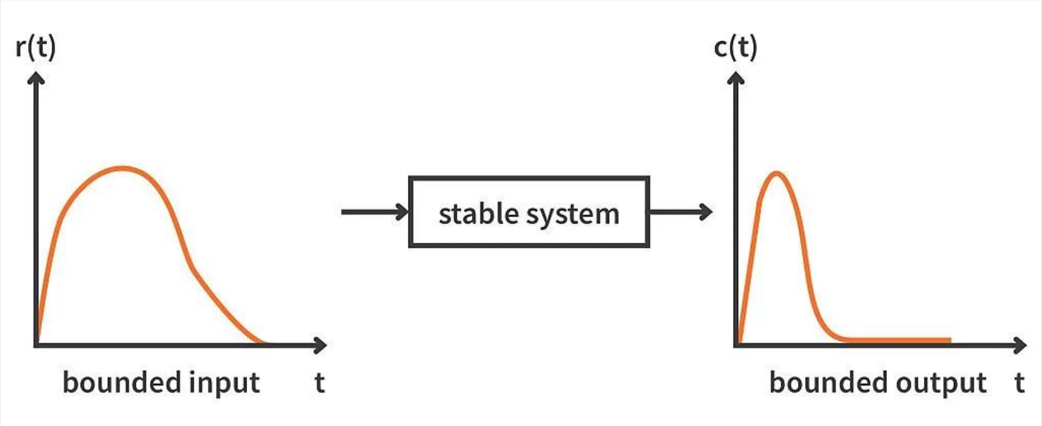
- A signal $x(t)$ is said to be **bounded** if there is a positive real number M such that
$$|x(t)| \leq M$$
for all $t \in [0, \infty)$.

- M is a bound of signal $x(t)$. If a signal is bounded, then its bounds are **not unique**.

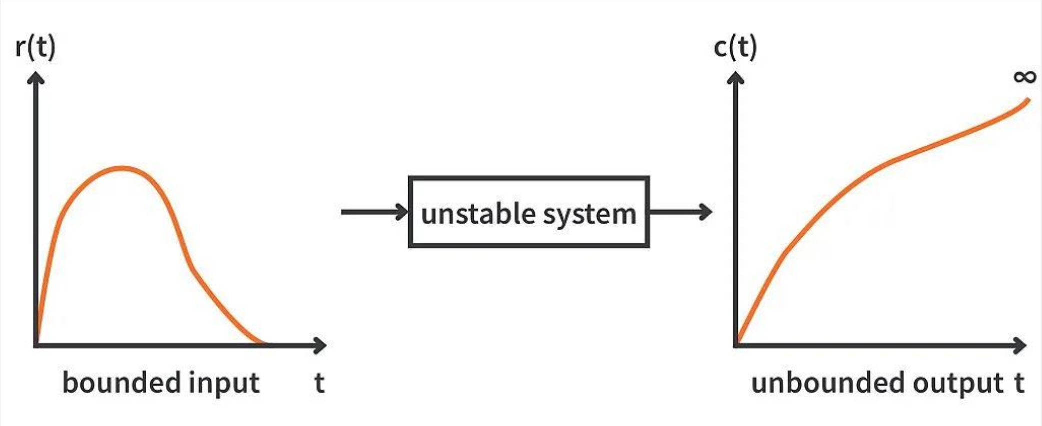
- The **smallest** bound, called the **peak magnitude** or **amplitude of the signal**, can be denoted by
$$\|x(t)\|_\infty = \sup_{t \in [0,\infty)} |x(t)|.$$

Concept of stable





For every bounded input signal, if the system response is also bounded, then that system is stable.



For any bounded input, if the system response is unbounded, then that system is unstable.

BIBO Stability

- **Definition** A system is said to be **bounded-input-bounded-output (BIBO) stable** if for each bounded input the corresponding output is bounded.

※ To show a system is **unstable**, we only need to find a **counterexample**, i.e., a bounded input produces an **unbounded output**.

※ To show a system is **stable**, we need to show the output is bounded for **every arbitrary** bounded input. **Hard!!**

BIBO Stability

- **Example** Determine the stability of $G(s) = \dfrac{1}{s+\gamma}$.
  - If $\gamma < 0$, let $u(t) = \sigma(t)$, which is obviously bounded. Then,

$$y(t) = \mathcal{L}^{-1}\left[\frac{1}{s\,(s+\gamma)}\right] = \mathcal{L}^{-1}\left[\frac{1}{\gamma}\left(\frac{1}{s} - \frac{1}{s+\gamma}\right)\right] = \frac{1}{\gamma}(1 - e^{-\gamma t})\sigma(t).$$

    When $t \to \infty$, $y(t) \to \infty$, hence the system is **unstable**.
  - If $\gamma = 0$, the system is simply an **integrator**. Let $u(t) = \sigma(t)$, then

$$y(t) = \mathcal{L}^{-1}\left[\frac{1}{s^2}\right] = t\sigma(t).$$

    When $t \to \infty$, $y(t) \to \infty$. **Unstable!!**

BIBO Stability

- **Example** Determine the stability of $G(s) = \dfrac{1}{s+\gamma}$.
  - If $\gamma > 0$, the impulse response of $G(s)$ is $g(t) = e^{-\gamma t}\sigma(t)$. Then,

$$|y(t)| = \left| \int_0^t g(\tau)u(t-\tau)d\tau \right| \leq \int_0^t |g(\tau)||u(t-\tau)|d\tau.$$

If $|u(t)| \leq M$ for all $t \in [0, \infty)$, then

$$|y(t)| \leq M \int_0^\infty |g(\tau)|d\tau = M \int_0^\infty e^{-\gamma\tau}d\tau = \frac{M}{\gamma}.$$

As long as $u(t)$ is bounded by M, then $y(t)$ is bounded by M/γ. **Stable!!**

Stability Criterion

▶ **Theorem** An LTI system with impulse response function $g(t)$ is stable if and only if

$$\int_0^\infty |g(t)|dt < \infty. \qquad (3.1)$$

Proof. Assume that the input $u(t)$ is bounded by M; then

$$|y(t)| = \left|\int_0^t g(\tau)u(t-\tau)d\tau\right| \le \int_0^t |g(\tau)||u(t-\tau)|d\tau \le M\int_0^\infty |g(\tau)|d\tau.$$

If (3.1) is satisfied, choose $N = M\int_0^\infty |g(\tau)|d\tau$, and then the input bounded by M generates the output bounded by N; i.e., the system is stable. This shows the sufficiency.

If (3.1) is not satisfied, $\int_0^\infty |g(t)|dt = \infty$. For any $M > 0$, choose t_0 so that

$$\int_0^{t_0} |g(\tau)|d\tau > M$$

and choose

$$u(t) = \begin{cases} 1, & \text{if } g(t_0 - t) \ge 0; \\ -1, & \text{if } g(t_0 - t) < 0 \end{cases}$$

for $0 \le t \le t_0$. Then,

$$|y(t_0)| = \left|\int_0^{t_0} g(t_0-\tau)u(\tau)d\tau\right| = \int_0^{t_0} |g(t_0-\tau)|d\tau = \int_0^{t_0} |g(\tau)|d\tau > M.$$

This shows that for an input $u(t)$ bounded by 1, the output can be arbitrarily large. Hence, the system is not stable. This proves the necessity. □

Stability Criterion

▶ **Theorem** An LTI system with impulse response function $g(t)$ is stable if and only if

$$\int_0^\infty |g(t)|dt < \infty.$$

- **Absolutely integrable** A signal is said to be **absolutely integrable** over an interval if the integral of the absolute value of the signal over the interval is **finite**.

$$\int |f|\, d\mu < \infty$$

- A linear system is stable if its impulse response is absolutely integrable over $[0, \infty)$.

Stability Criterion

- **Example** Determine the stability of $G(s) = \dfrac{1}{s+\gamma}$ by testing $g(t)$.

  - The impulse response of $G(s)$ is

$$g(t) = e^{-\gamma t}\sigma(t).$$

This gives

$$\int_0^\infty |g(t)|dt = \int_0^\infty e^{-\gamma t}dt = \begin{cases} \dfrac{1}{\gamma} & \text{if } \gamma > 0 \\ \infty & \text{if } \gamma \leq 0. \end{cases}$$

The system is **stable** if $\gamma > 0$ and **unstable** if $\gamma \leq 0$. **Same results!!**

Stability Criterion

▶ **Theorem** A system with transfer function $G(s)$ is **stable** if and only if $G(s)$ is **proper** and **all poles** of $G(s)$ have **negative real parts**.

***Proof*.** If the system is stable, then

$$\int_0^\infty |g(t)|dt < \infty.$$

This implies that for all s with $\mathrm{Re}(s) \geq 0$,

$$|G(s)| = \left|\int_0^\infty g(t)e^{-st}dt\right| \leq \int_0^\infty |g(t)|dt < \infty,$$

i.e., $G(s)$ is bounded in the right half of the complex plane. Therefore $G(s)$ has to be proper and free of poles in the right half of the complex plane.

Stability Criterion

(Continued)

If $G(s)$ is proper and all poles of $G(s)$ have negative real parts, then $G(s)$ has the following partial fractional expansion:

$$G(s) = \sum_i G_i(s)$$

where $G_i(s)$ takes three possible forms:

$$G_i(s) = a \quad \text{or} \quad \frac{b}{(s+\gamma)^m} \quad \text{or} \quad \frac{q(s)}{[(s+\rho)^2+\omega^2]^l}$$

where $\gamma > 0$, $\rho > 0$, and $q(s)$ is an lth order polynomial of s. Hence,

$$g_i(t) = \mathcal{L}^{-1}[G_i(s)] = a\delta(t) \text{ or } e^{-\gamma t}p(t) \text{ or } e^{-\rho t}[p_1(t)\cos\omega t + p_2(t)\sin\omega t]$$

where $p(t), p_1(t), p_2(t)$ are polynomials of t. Since for each of these possibilities

$$\int_0^\infty |g_i(t)|dt < \infty$$

following the fact that the exponential functions $e^{-\gamma t}$ and $e^{-\rho t}$ converge to zero much faster than the speed in which $p(t), p_1(t), p_2(t)$ go to infinity, it follows that

$$\int_0^\infty |g(t)|dt \leq \sum_i \int_0^\infty |g_i(t)|dt < \infty.$$

This shows that the system is stable. $\square$

Stability Criterion

▶ **Example** Determine the stability of $G(s) = \dfrac{1}{s+\gamma}$ by checking the sign of real part of the pole.

- The pole of $G(s)$ is $s_1 = -\gamma$, and only has **real part**.
  - If $\gamma \leq 0$, then $s_1 \geq 0$, hence the system is **unstable**.
  - If $\gamma > 0$, then $s_1 < 0$. **Stable!!**.

PID Systems

- **Properness** of the transfer function for the stability of a system is important. The system with a **nonproper** transfer function is **unstable**.
- **Ideal differentiator** $G(s) = s$ is **unstable**.
- **Ideal integrable** $G(s) = \dfrac{1}{s}$ is **unstable**.
- **Ideal PID system** A system with transfer function

$$G(s) = K_P + K_I \frac{1}{s} + K_D s,$$

known as a PID system, is **unstable**.

- **Practical differentiator** A transfer function $G(s) = \dfrac{s}{Ts+1}$ with a very **small positive** T.

Polynomial Stability

- The poles of a transfer function $G(s)$ are the **roots** of its denominator polynomial.

- Determine the **stability of an LTI system**

  $\iff$ check the **root positions** of a polynomial

  $\iff$ determine the **stability of a polynominal**.

▶ **Definition** A polynomial is said to be stable if all of its roots have negative real parts.

▶ Is it possible to determine the stability of a polynomial **without computing the roots**?

Polynomial Stability

- Consider a polynomial:

$$a(s) = a_0 s^n + a_1 s^{n-1} + \cdots + a_n, \qquad a_0 > 0.$$

▶ **Theorem** If $a(s)$ is stable, then $a_i > 0$, $i = 1, \ldots, n$.

※ This theorem only gives a **necessary condition** for stability.

Polynomial Stability

- Consider a polynomial:

$$a(s) = a_0 s^n + a_1 s^{n-1} + \cdots + a_n, \qquad a_0 > 0.$$

▶ **Theorem** If $a(s)$ is stable, then $a_i > 0$, $i = 1, \ldots, n$.

※ This theorem only gives a **necessary condition** for stability.

▶ **Example** Consider a polynomial

$$a(s) = s^3 + s^2 + 4s + 30 = (s+3)(s^2 - 2s + 10).$$

This polynomial has all **positive** coefficients but it is **unstable**. Its roots are -3 and $1 \pm 3j$.

Construct Routh Table

- The **first two rows** come **directly** from the coefficients of $a(s)$.
- Each of the **other rows** is computed from its **two preceding rows** as
$$r_{ij} = -\frac{1}{r_{(i-1)0}} \det \begin{bmatrix} r_{(i-2)0} & r_{(i-2)(j+1)} \\ r_{(i-1)0} & r_{(i-1)(j+1)} \end{bmatrix}.$$
- Whenever $r_{(i-1)(j+1)}$ is **missing**, simply let $r_{(i-1)(j+1)} = 0$, then $r_{ij} = r_{(i-2)(j+1)}$.

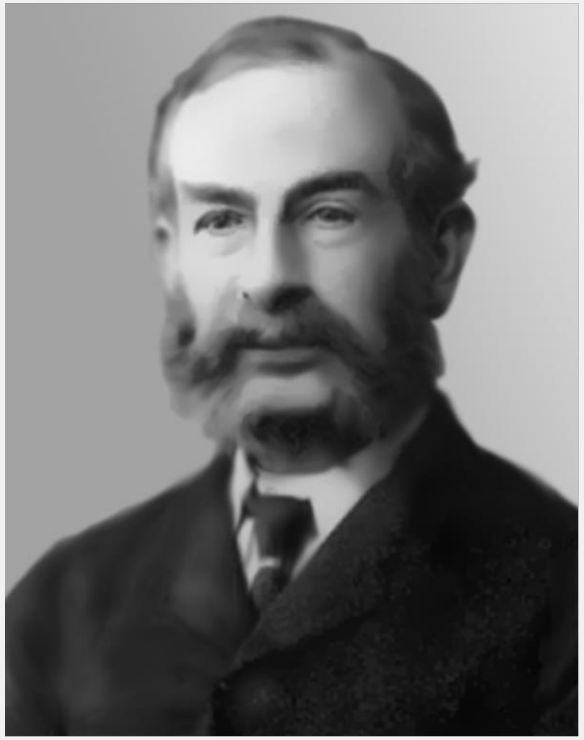
s^n	$r_{00} = a_0$	$r_{01} = a_2$	$r_{02} = a_4$	$r_{03} = a_6$	$\cdots$
s^{n-1}	$r_{10} = a_1$	$r_{11} = a_3$	$r_{12} = a_5$	$r_{13} = a_7$	$\cdots$
s^{n-2}	r_{20}	r_{21}	r_{22}	r_{23}	$\cdots$
s^{n-3}	r_{30}	r_{31}	r_{32}	r_{33}	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
s^2	$r_{(n-2)0}$	$r_{(n-2)1}$			
s^1	$r_{(n-1)0}$				
s^0	r_{n0}				

Routh Stability Criterion

▶ **Routh Stability Criterion**

The following three statements are equivalent:

1. $a(s)$ is **stable**.
2. All elements of the Routh table are **positive**, i.e., $r_{ij} > 0$, $i = 0, 1, \ldots, n$, $j = 0, 1, \ldots, \lfloor \frac{n-i}{2} \rfloor$.
3. All elements in the **first column** of the Routh table are **positive**, i.e., $r_{i0} > 0$, $i = 0, 1, \ldots, n$.



Edward John Routh (1831 - 1907)

Routh Stability Criterion

▶ **Example** Determine the stability of $a(s) = s^4 + 10s^3 + 35s^2 + 50s + 24$.

- **Construct the Routh table**

$$
\begin{array}{c|ccc}
\hline
s^4 & 1 & 35 & 24 \\
s^3 & 10 & 50 & \\
s^2 & r_{20} = 30 & r_{21} = 24 & \\
s^1 & r_{30} = 42 & & \\
s^0 & r_{40} = 24 & &
\end{array}
$$

$$r_{20} = \frac{-1}{10} \det \begin{bmatrix} 1 & 35 \\ 10 & 50 \end{bmatrix} = 30, \qquad r_{21} = \frac{-1}{10} \det \begin{bmatrix} 1 & 24 \\ 10 & \mathbf{0} \end{bmatrix} = 24,$$

$$r_{30} = \frac{-1}{30} \det \begin{bmatrix} 10 & 50 \\ 30 & 24 \end{bmatrix} = 42, \qquad r_{40} = \frac{-1}{42} \det \begin{bmatrix} 30 & 24 \\ 42 & \mathbf{0} \end{bmatrix} = 24.$$

Routh Stability Criterion

▶ **Example** Determine the stability of $a(s) = s^4 + 10s^3 + 35s^2 + 50s + 24$.

- **Construct the Routh table**

$$
\begin{array}{c|ccc}
\hline
s^4 & 1 & 35 & 24 \\
s^3 & 10 & 50 & \\
s^2 & r_{20} = 30 & r_{21} = 24 & \\
s^1 & r_{30} = 42 & & \\
s^0 & r_{40} = 24 & &
\end{array}
$$

- Since every element in the **first column** of the table is **positive**, the polynominal is **stable**.

Routh Stability Criterion

- **Example** Determine the stability of $a_1(s) = s^3 + 2s^2 + 3s + 10$.

  - **Construct the Routh table**

s^3	1	3
s^2	2	10
s^1	$\mathbf{-2}$	
s^0	10	

  - Since the **first column** contains $\mathbf{-2}$, the polynominal is **unstable**.

Routh Stability Criterion

- **Example** Determine the stability of $a_2(s) = s^5 + 10s^4 + 35s^3 + 50s^2 + 24s + 300$.
  - **Construct the Routh table**

s^5	1	35	24
s^4	10	50	300
s^3	$r_{20} = 30$	$r_{21} = \mathbf{-6}$	

  - As soon as the **negative entry** appears, the polynominal is **unstable**.

Routh Stability Criterion

▶ **Example** Determine the stability of $a_3(s) = s^4 + 3s^3 + 3s^2 + 3s + 2$.

- **Construct the Routh table**

s^4	1	3	2
s^3	3	3	
s^2	2	2	
s^1	$r_{30} = \mathbf{0}$		
s^0	r_{40}		

- Since $r_{30} = 0$, r_{40} cannot be computed. The polynominal is **unstable**.

Routh Stability Criterion

- **Example** Determine the value of K such that $a(s) = s^4 + 2s^3 + 4s^2 + 4s + K$ is stable.

4-k > 0 ⇒ k < 4
and
k > 0

  - **Construct the Routh table**

s^4	1	4	K
s^3	2	4	
s^2	2	K	
s^1	$4-K$		
s^0	K		

1	4	k
2	4	0
2	k	0
4-k	0	0
K		

  - $a(s)$ is stable if $K > 0$ and $4 - K > 0$. Hence the polynominal $a(s)$ is **stable** iff $0 < K < 4$.

0 < K < 4

Robust Stability

- If the **coefficients** of a polynomial $a(s)$ are in **certain ranges**, how can we determine the stability of $a(s)$?

▶ **Interval polynomial** Consider a set of polynomials:

$$\mathcal{P} = \{a(s) = a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n : a_i \in [\underline{a}_i, \overline{a}_i]\}.$$

This set of polynomials is often called an **interval polynomial**.

Kharitonov Theorem

- **Interval polynomial** Consider a set of polynomials:

$$\mathcal{P} = \{a(s) = a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n : a_i \in [\underline{a}_i, \overline{a}_i]\}.$$

This set of polynomials is often called an **interval polynomial**.

LLHH , LHHL , HLLH , HHLL

- **Kharitonov Theorem** All members of $\mathcal{P}$ are stable if and only if the following **four polynomials** are stable:

$$a_1(s) = \underline{a}_0 s^n + \underline{a}_1 s^{n-1} + \overline{a}_2 s^{n-2} + \overline{a}_3 s^{n-3} + \underline{a}_4 s^{n-4} + \cdots$$

$$a_2(s) = \underline{a}_0 s^n + \overline{a}_1 s^{n-1} + \overline{a}_2 s^{n-2} + \underline{a}_3 s^{n-3} + \underline{a}_4 s^{n-4} + \cdots$$

$$a_3(s) = \overline{a}_0 s^n + \underline{a}_1 s^{n-1} + \underline{a}_2 s^{n-2} + \overline{a}_3 s^{n-3} + \overline{a}_4 s^{n-4} + \cdots$$

$$a_4(s) = \overline{a}_0 s^n + \overline{a}_1 s^{n-1} + \underline{a}_2 s^{n-2} + \underline{a}_3 s^{n-3} + \overline{a}_4 s^{n-4} + \cdots .$$

Kharitonov Theorem

- **Example** Is every polynomial in this set stable?

$\mathcal{P} = \{s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 : a_1 \in [3,4], a_2 \in [3,4], a_3 \in [2,3], a_4 \in [0.5,1]\}.$

- **Kharitonov Theorem**

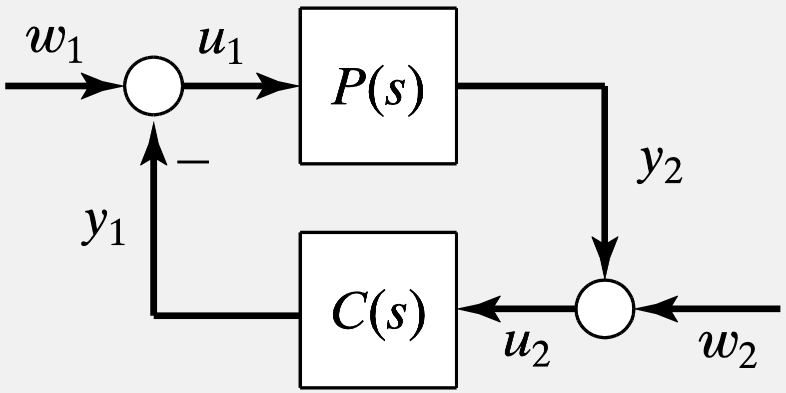
$$\begin{aligned} a_1(s) &= s^4 + 3s^3 + 4s^2 + 3s + 0.5 \\ a_2(s) &= s^4 + 4s^3 + 4s^2 + 2s + 0.5 \\ a_3(s) &= s^4 + 3s^3 + 3s^2 + 3s + 1 \\ a_4(s) &= s^4 + 4s^3 + 3s^2 + 2s + 1. \end{aligned}$$

- By using Routh Stability Criterion, all four polynomials are stable. Hence, the original $\mathcal{P}$ is **stable**.

Stability of Closed-loop Systems

- One fundamental purpose of control is to design a controller so that a closed-loop system is **stable** and possibly satisfies other **performance specifications**.
- **Two basic control problem**
  - Feedback systems for **stabilization**
  - Feedback systems for **regulation**

Feedback Systems for Stabilization



A system is internally stable if for all initial conditions, and all bounded signals injected at any place in the system, all states remain bounded for all future time.

▶ **Definition** The closed-loop system is said to be **internally stable** if the **8** transfer functions from w_i, $i = 1, 2$, to u_j and y_j, $j = 1, 2$, are stable.

- **Gang of four** **Essential!!**

$$\boxed{\frac{1}{1+P(s)C(s)}, \quad \frac{C(s)}{1+P(s)C(s)}, \quad \frac{P(s)}{1+P(s)C(s)}, \quad \frac{P(s)C(s)}{1+P(s)C(s)}.}$$

Gang of Four Matrix

- Let

$$P(s) = \frac{b(s)}{a(s)} \quad \text{and} \quad C(s) = \frac{q(s)}{p(s)}$$

where $a(s)$ and $b(s)$ are **coprime** polynomials and so are $p(s)$ and $q(s)$.

- **Gang of four matrix**

$$GoF(s) = \begin{bmatrix} \frac{1}{1+P(s)C(s)} & \frac{C(s)}{1+P(s)C(s)} \\ \frac{P(s)}{1+P(s)C(s)} & \frac{P(s)C(s)}{1+P(s)C(s)} \end{bmatrix} = \frac{\begin{bmatrix} a(s)p(s) & a(s)q(s) \\ b(s)p(s) & b(s)q(s) \end{bmatrix}}{a(s)p(s)+b(s)q(s)}.$$

Feedback Systems for Stabilization

- **Theorem** Let $P(s) = \dfrac{b(s)}{a(s)}$ and $C(s) = \dfrac{q(s)}{p(s)}$ be **proper** transfer functions and assume $1 + P(\infty)C(\infty) \neq 0$. These statements are **equivalent**:

  1. The closed-loop system is internally stable.
  2. The polynomial $a(s)p(s) + b(s)q(s)$ is stable.
  3. There is **no unstable pole/zero cancellation** in forming the product $P(s)C(s)$ and any one of the **gang of four** is stable.

Characteristic Polynomial

- **Definition** For a feedback system for stabilization with $P(s) = \dfrac{b(s)}{a(s)}$ and $C(s) = \dfrac{q(s)}{p(s)}$, the polynomial $c(s) = a(s)p(s) + b(s)q(s)$ is called its **characteristic polynomial**.

Internal Stability Examples

- **Example** Let $P(s) = \dfrac{s-1}{s(s-2)}$ and $C(s) = \dfrac{33s-6}{s-25}$. Both $P(s)$ and $C(s)$ are **unstable**, but the closed-loop characteristic polynomial

$$s(s-2)(s-25) + (s-1)(33s-6) = s^3 + 6s^2 + 11s + 6$$

is **stable**. Hence, the closed-loop system is **stable**.

- **Example** Let $P(s) = \dfrac{1}{(s+1)^3}$ and $C(s) = 10$. Both $P(s)$ and $C(s)$ are **stable**, but the closed-loop characteristic polynomial

$$(s+1)^3 + 10 = s^3 + 3s^2 + 3s + 11$$

is **unstable**. Hence, the closed-loop system is **unstable**.

$$\begin{array}{cc} 1 & 3 \\ 3 & 11 \\ -\frac{2}{3} & \end{array}$$

$\Rightarrow$ unstable

Internal Stability Examples

- **Example** Let $P(s) = \dfrac{1}{s-1}$ and $C(s) = \dfrac{s-1}{s+1}$. Then

$$\frac{1}{1+P(s)C(s)} = \frac{s+1}{s+2}, \qquad \frac{P(s)}{1+P(s)C(s)} = \frac{s+1}{{\color{red}(s-1)}(s+2)},$$
$$\frac{C(s)}{1+P(s)C(s)} = \frac{s-1}{s+2}, \qquad \frac{P(s)C(s)}{1+P(s)C(s)} = \frac{1}{s+2}.$$

Hence, the closed-loop system is **unstable** even though **three out of four** different closed-loop transfer functions are **stable**.

Unstable cancellations!!

Internal Stability Examples

- **Example** Let $P(s) = \dfrac{s-2}{s+1}$ and $C(s) = \dfrac{1}{s-2}$. Then

$$\frac{1}{1+P(s)C(s)} = \frac{s+1}{s+2}, \qquad \frac{P(s)}{1+P(s)C(s)} = \frac{s-2}{s+2},$$

$$\frac{C(s)}{1+P(s)C(s)} = \frac{(s+1)}{(s+2)(s-2)}, \qquad \frac{P(s)C(s)}{1+P(s)C(s)} = \frac{1}{s+2}.$$

Hence, the closed-loop system is **unstable**.

Unstable cancellations!!

Internal Stability Examples

- **Example** Let $P(s) = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ and $C(s) = K_D s + K_P$. Such a controller is called a **proportional-derivative (PD)** controller. Then

$$\frac{P(s)C(s)}{1 + P(s)C(s)} = \frac{(K_D s + K_P)\omega_n^2}{s^2 + (2\zeta\omega_n + K_D\omega_n^2)s + (1 + K_P)\omega_n^2}$$

which can be made stable by designing K_D and K_P, but

$$\frac{C(s)}{1 + P(s)C(s)} = \frac{(s^2 + 2\zeta\omega_n s + \omega_n^2)(K_D s + K_P)}{s^2 + (2\zeta\omega_n + K_D\omega_n^2)s + (1 + K_P)\omega_n^2}$$

which is **unstable** as long as $K_D \neq 0$ since it is not bounded at $s \to \infty$.

- **Practical differentiator** $C(s) = K_D \dfrac{s}{Ts + 1} + K_P$ with a **small** $T > 0$.

Internal Stability Examples

- **Example** Let $P(s) = \dfrac{s+2}{s+1}$ and $C(s) = -1$. Then

$$\frac{P(s)}{1+P(s)C(s)} = -s-2, \quad \frac{C(s)}{1+P(s)C(s)} = s+1. \qquad \textbf{Nonproper!!}$$

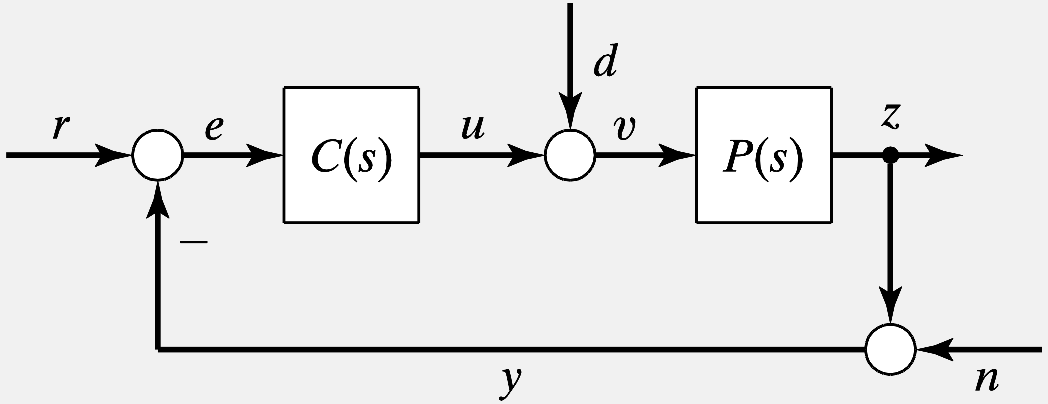
Hence, the closed-loop system is **unstable**.

- $a(s)p(s) + b(s)q(s) = +1$ is stable and both $P(s)$ and $C(s)$ are proper, but the **order** of the characteristic polynominal **drops** blow that of $a(s)p(s)$ or $b(s)q(s)$.

These examples show that several things can go wrong if we follow our intuition to check only one of the possible transfer functions or check only the polynomial a(s)p(s) + b(s)q(s) to decide closed-loop stability.

In general, in order to know the stability of the closed-loop system, it is not sufficient to check carelessly the stability of only one transfer function among the four different transfer functions. Nor is it sufficient to check carelessly the stability of polynomial a(s)p(s) + b(s)q(s).

Unity Feedback Systems



- **Example** Consider the unity feedback system with

$$P(s) = \frac{10}{(s+1)(s+2)}.$$

$10k_p s + 10k_I + s(s+1)(s+2)$

$s^3 + 3s^2 + 2s + 10k_p s + 10k_I$

Let $C(s)$ be a **proportional-integral (PI)** controller, i.e.,

$$C(s) = K_P + \frac{K_I}{s}.$$

1	$2+10k_p$
3	$10k_I$
$\frac{6+30k_p-10k_I}{3}$	0
$10k_I$	

Find the **ranges** of K_P and K_I so that the closed-loop system is stable.

Unity Feedback Systems

- **Example** Since $C(s)$ is proper and $P(s)$ is strictly proper, we only need to check the stability of the closed-loop characteristic polynomial:

$$d(s) = s^3 + 3s^2 + (2 + 10K_P)s + 10K_I.$$

Construct the Routh table:

s^3	1	$(2 + 10K_P)$
s^2	3	$10K_I$
s^1	$\dfrac{3(2 + 10K_P) - 10K_I}{3}$	
s^0	$10K_I$	

$k_I > 0$

$6 + 30k_P - 10k_I > 0$

$k_P > \dfrac{10k_I - 6}{30}$

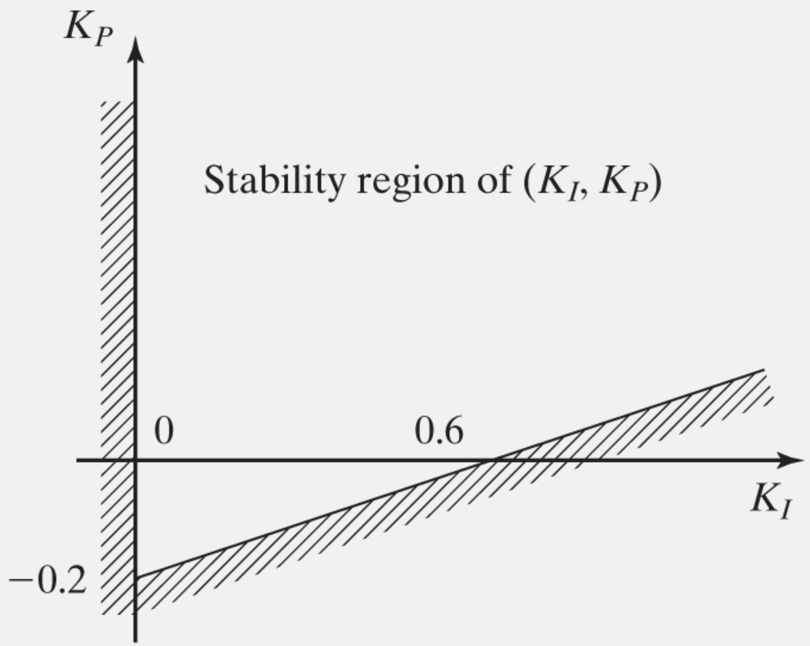
$k_P > \dfrac{5k_I - 3}{15}$

Unity Feedback Systems

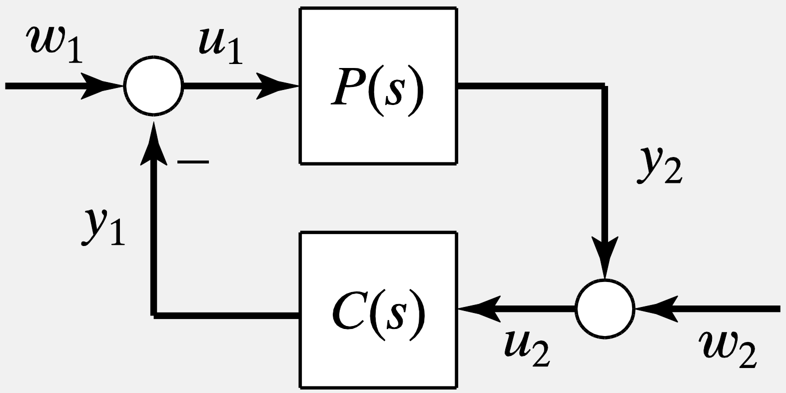
- **Example** By Routh criterion, we need

$$\frac{3(2+10K_P)-10K_I}{3} > 0, \quad 10K_I > 0 \Longleftrightarrow 3+15K_P-5K_I > 0, \quad K_I > 0.$$

The stability region of K_I and K_P is plotted.



Pole Placement Design

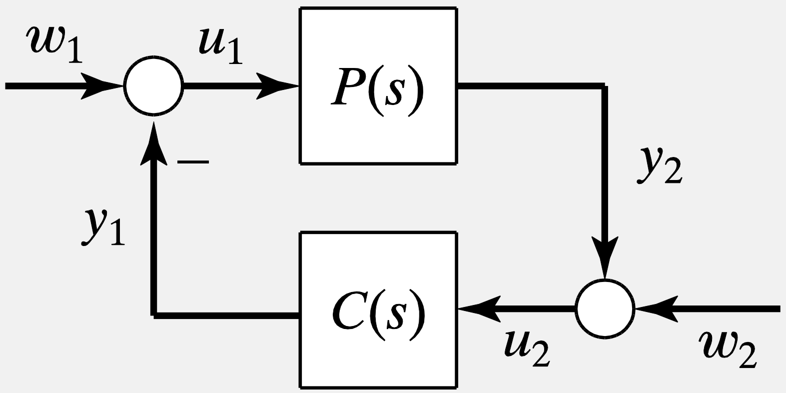


- **Stabilizing controller desgin:**

  Given the plant $P(s)$, design the controller $C(s)$ so that the closed-loop system is **internally stable**.

  - Proper controller
  - Strictly proper controller

Pole Placement Design



- **Idea of pole placement:**

  Design $C(s)$ so that

  the **poles** of the closed-loop system

  $\Longleftrightarrow$ the **roots** of the characteristic polynomial

  are placed in the **prespecified desirable locations**.

Proper Controllers

- Let a plant be given by

$$P(s) = \frac{b(s)}{a(s)} = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{a_0 s^n + a_1 s^{n-1} + \cdots + a_n}, \quad a_0 \neq 0,$$

where $a(s)$ and $b(s)$ are coprime.

- Consider **proper** controllers:

$$C(s) = \frac{q(s)}{p(s)} = \frac{q_0 s^m + q_1 s^{m-1} + \cdots + q_m}{p_0 s^m + p_1 s^{m-1} + \cdots + p_m}, \quad p_0 \neq 0,$$

where $p(s)$ and $q(s)$ are coprime.

Proper Controllers

1. Matrix method
2. Compare coefficients

- The closed-loop **characteristic polynomial** is

known

$$\boxed{c(s) := a(s)p(s) + b(s)q(s) = c_0 s^{n+m} + c_1 s^{n+m-1} + \cdots + c_{n+m}.}$$

Given $a(s), b(s), c(s)$ and **unknown** $p(s), q(s)$, it is also called a polynomial **Diophantine equation**.

- **Stabilizing controller design:**
  - Specify an $(n+m)$th order desired stable $c(s)$.
  - Choose $p(s)$ and $q(s)$ so that the Diophantine equation is satisfied.
  - Set $C(s) = \dfrac{q(s)}{p(s)}$.

Pole Placement Design

- Define two $(n+m)\times m$ **Toeplitz matrices**:

$$\boldsymbol{T}(a(s),m) = \overbrace{\begin{bmatrix} a_0 & 0 & \cdots & 0 \\ a_1 & a_0 & \ddots & \vdots \\ \vdots & a_1 & \ddots & 0 \\ a_n & \vdots & \ddots & a_0 \\ 0 & a_n & \ddots & a_1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix}}^{m \text{ columns}}, \quad \boldsymbol{T}(b(s),m) = \overbrace{\begin{bmatrix} b_0 & 0 & \cdots & 0 \\ b_1 & b_0 & \ddots & \vdots \\ \vdots & b_1 & \ddots & 0 \\ b_n & \vdots & \ddots & b_0 \\ 0 & b_n & \ddots & b_1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_n \end{bmatrix}}^{m \text{ columns}}.$$

- The elements in each of their diagonals are equal.

Pole Placement Design

- Define an $(n+m) \times 2m$ **Sylvester's resultant matrix**

$$\boldsymbol{S}(a(s), b(s), m) = \begin{bmatrix} \boldsymbol{T}(a(s), m) & \boldsymbol{T}(b(s), m) \end{bmatrix}.$$

  - $\boldsymbol{S}(a(s), b(s), n)$ is nonsingular $\Longleftrightarrow$ $a(s)$ and $b(s)$ are coprime.
  - $\boldsymbol{S}(a(s), b(s), m)$ has full column rank when $m < n$ and has full row rank when $m > n$.

Pole Placement Design

- Recall the Diophantine equation is

$$a(s)p(s) + b(s)q(s) = c_0 s^{n+m} + c_1 s^{n+m-1} + \cdots + c_{n+m}.$$

- Compare the **coefficients** of both sides of this equation: Also compare of coeffecients

$$\begin{bmatrix} a_0 & 0 & \cdots & 0 & b_0 & 0 & \cdots & 0 \\ a_1 & a_0 & \ddots & \vdots & b_1 & b_0 & \ddots & \vdots \\ \vdots & a_1 & \ddots & 0 & \vdots & b_1 & \ddots & 0 \\ a_n & \vdots & \ddots & a_0 & b_n & \vdots & \ddots & b_0 \\ 0 & a_n & \ddots & a_1 & 0 & b_n & \ddots & b_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n & 0 & 0 & \cdots & b_n \end{bmatrix} \begin{bmatrix} \left.\begin{matrix} p_0 \\ \vdots \\ p_m \end{matrix}\right\} m+1 \\ q_0 \\ \vdots \\ q_m \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n+m-1} \\ c_{n+m} \end{bmatrix}.$$

$\boldsymbol{S}(a(s), b(s), m+1)$

$$\begin{bmatrix} a_0 & \vdots \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_m \end{bmatrix} + \begin{bmatrix} b_0 & \vdots \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ \vdots \\ q_m \end{bmatrix}$$

$$\begin{bmatrix} a_0 & & & \vdots & b_0 \\ a_1 & a_0 & & \vdots & b_1 \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & & & \vdots & \vdots \\ 0 & & a_0 & \vdots & 0 \end{bmatrix} \begin{bmatrix} p_0 \\ \vdots \\ p_m \\ q_0 \\ \vdots \\ q_m \end{bmatrix} = \begin{bmatrix} c_0 \\ \vdots \\ c_{n+m} \end{bmatrix}$$

$\longleftrightarrow$ $m+1$ $\qquad$ $\longleftrightarrow$ $m+1$

$n+m+1 = 2m+2$

$n = m+1$

Pole Placement Design

$$\boldsymbol{S}(a(s), b(s), m+1) \begin{bmatrix} p_0 \\ \vdots \\ p_m \\ q_0 \\ \vdots \\ q_m \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n+m-1} \\ c_{n+m} \end{bmatrix}.$$

- If $c(s)$ is arbitrarily chosen, when

  col row

  - $m+1 < n$: $\boldsymbol{S}(a(s), b(s), m+1)$ has full column rank. **No solution!!**
  - $m+1 = n$: $\boldsymbol{S}(a(s), b(s), m+1)$ is nonsingular. **Unique solution.**
  - $m+1 > n$: $\boldsymbol{S}(a(s), b(s), m+1)$ is a wide matrix. **Infinitely many solutions.**

Pole Placement Design

$$\boldsymbol{S}(a(s), b(s), m+1)\begin{bmatrix} p_0 \\ \vdots \\ p_m \\ q_0 \\ \vdots \\ q_m \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n+m-1} \\ c_{n+m} \end{bmatrix}.$$

- **Summary** There exists an mth order, **proper** controller so that the closed-loop poles are **arbitrarily assigned** to a set of $n+m$ complex numbers $\Longleftrightarrow m \geq n-1$.

Pole Placement Design

- **Example** Let

$$P(s) = \frac{s-1}{s(s-2)}.$$

q

p

$ap + bq = c$

$a\,s(s-2) + b(s-1)$

$= s^3 + 6s^2 + 11s + 6$

Design a **first-order** controller so that the closed-loop poles are

$$-1, \quad -2, \quad -3.$$

$$\Longleftrightarrow \quad c(s) = (s+1)(s+2)(s+3) = s^3 + 6s^2 + 11s + 6.$$

$$C = \frac{q_0 s + q_1}{p_0 s + p_1}$$

Pole Placement Design

- **Example** Denote

$$C(s) = \frac{q_0 s + q_1}{p_0 s + p_1}.$$

Then the Diophantine equation is

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 0 & -2 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ q_0 \\ q_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 11 \\ 6 \end{bmatrix}.$$

$$\Longrightarrow p_0 = 1, \quad p_1 = -25, \quad q_0 = 33, \quad q_1 = -6.$$

$$\Longrightarrow C(s) = \frac{33s - 6}{s - 25}.$$

Stable controller?

non stable

Another method using Comparing Coefficient

- **Example** Let

$$P(s) = \frac{s-1}{s(s-2)}.$$

Design a **first-order** controller so that the closed-loop poles are

$$-1, \quad -2, \quad -3.$$

$$\iff c(s) = (s+1)(s+2)(s+3) = s^3 + 6s^2 + 11s + 6.$$

- **Another method** Compare the coefficients of

$$s(s-2)(p_0 s + p_1) + (s-1)(q_0 s + q_1) = s^3 + 6s^2 + 11s + 6.$$

$$\implies p_0 = 1, \quad -2p_0 + p_1 + q_0 = 6, \quad -2p_1 - q_0 + q_1 = 11, \quad -q_1 = 6.$$

Same results.

Pole Placement Design

- We may be able to place the closed-loop poles to **certain particular positions** by using controllers with an **order less than** $n-1$.

▶ **Example** Let

$$P(s) = \frac{1}{s(s+2)}.$$

- **Zeroth order controller** Using $C(s) = q_0/p_0$, place the closed-loop poles to

$$-1 \; and -1 \Longleftrightarrow c(s) = s(s+2)p_0 + q_0 = (s+1)^2.$$
$$\Longrightarrow p_0 = q_0 = 1, \text{i.e.}, C(s) = 1.$$

This particular $c(s)$ makes the equations **not independent**.

Pole Placement Design

- **First-order controller** Using

$$C(s) = \frac{q_0 s + q_1}{p_0 s + p_1},$$

place the closed-loop poles to $-1, -1, -2 \iff$ Diophantine equation is

$$c(s) = s(s+2)(p_0 s + p_1) + (q_0 s + q_1) = (s+1)^2(s+2).$$

$$\implies p_0 = 1, \quad p_1 = 2, \quad q_0 = 1, \quad p_1 = 2.$$

$$\implies C(s) = \frac{s+2}{s+2} = 1.$$

Same results.

Note: As shown in these two examples, the lowest order of a stabilizing controller for an nth order plant might be lower than n−1, although the lowest order of controllers for arbitrary pole placement is n − 1.

Strictly Proper Controllers

- A **strictly proper** controller has the form

$$C(s) = \frac{q(s)}{p(s)} = \frac{q_1 s^{m-1} + \cdots + q_m}{p_0 s^m + p_1 s^{m-1} + \cdots + p_m}$$

  where $p(s)$ and $q(s)$ are coprime and $p_0 \neq 0$.

- The closed-loop characteristic polynomial is still

$$c(s) := a(s)p(s) + b(s)q(s) = c_0 s^{n+m} + c_1 s^{n+m-1} + \cdots + c_{n+m}.$$

Pole Placement Design

- **The Diophantine equation:**

$$\boxed{\begin{cases} p_0 = c_0/a_0 \\ \boldsymbol{S}(a(s), b(s), m) \begin{bmatrix} p_1 \\ \vdots \\ p_m \\ q_1 \\ \vdots \\ q_m \end{bmatrix} = \begin{bmatrix} c_1 - a_1 c_0/a_0 \\ \vdots \\ c_n - a_n c_0/a_0 \\ c_{n+1} \\ \vdots \\ c_{n+m} \end{bmatrix}. \end{cases}}$$

- If $c(s)$ is arbitrarily chosen, when
  - $m = n$: **unique solution**.
  - $m > n$: **infinitely many solutions**.

Pole Placement Design

- **The Diophantine equation:**

$$\begin{cases} p_0 = c_0/a_0 \\ \boldsymbol{S}(a(s), b(s), m) \begin{bmatrix} p_1 \\ \vdots \\ p_m \\ q_1 \\ \vdots \\ q_m \end{bmatrix} = \begin{bmatrix} c_1 - a_1 \boxed{c_0/a_0} \\ \vdots \\ c_n - a_n c_0/a_0 \\ c_{n+1} \\ \vdots \\ c_{n+m} \end{bmatrix} . \end{cases}$$

=1 most time!

- **Summary** There exists an mth order, **strictly proper** controller so that the closed-loop poles are **arbitrarily assigned** to a set of $n+m$ complex numbers $\iff m \geq n$.

Pole Placement Design

- **Example** Let

$$P(s) = \frac{s-1}{s(s-2)}.$$

Design a **second-order**, **strictly proper** controller so that the closed-loop poles are

$$-1 \pm j1, \quad -2 \pm j2.$$

$$\iff c(s) = (s^2+2s+2)(s^2+4s+8) = s^4+6s^3+18s^2+24s+16.$$

Pole Placement Design

- **Example** Denote

$$C(s) = \frac{q_1 s + q_2}{p_0 s^2 + p_1 s + p_2}.$$

Then the Diophantine equation is

$$\begin{cases} p_0 = 1 \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 0 & -2 & -1 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 6-(-2) \\ 18-0 \\ 24 \\ 16 \end{bmatrix}. \end{cases}$$

$$\Longrightarrow p_0 = 1, \quad p_1 = 8, \quad p_2 = -74, \quad q_1 = 108, \quad q_2 = -16.$$

$$\Longrightarrow C(s) = \frac{108s - 16}{s^2 + 8s - 74}.$$

Unstable controller

Stable closed-loop system

Pole Placement Design

- **Another method** Compare the coefficients of

$$s(s-2)(p_0 s^2 + p_1 s + p_2) + (s-1)(q_1 s + q_2) = s^4 + 6s^3 + 18s^2 + 24s + 16.$$

$$\begin{aligned} \Longrightarrow\ & p_0 = 1, \quad -2p_0 + p_1 = 6, \quad -2p_1 + p_2 + q_1 = 18, \\ & -2p_2 - q_1 + q_2 = 24, \quad -q_2 = 16. \\ \Longrightarrow\ & p_0 = 1, \quad p_1 = 8, \quad p_2 = -74, \quad q_1 = 108, \quad q_2 = -16. \end{aligned}$$

Same results.

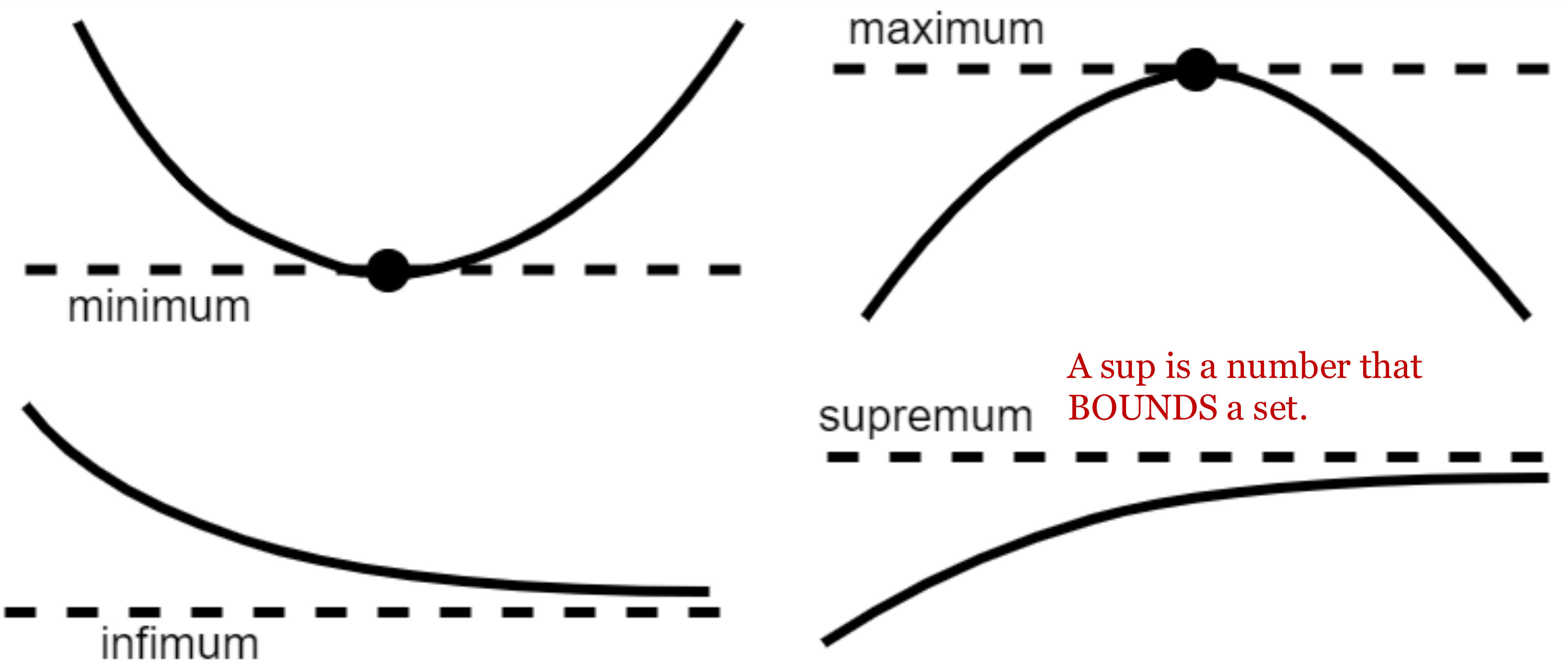
Review & Exercise

- What’s bounded signal?
- What’s BIBO stability?
- What’s the stability criterion? Two.
- What’s “properness”?
- What’s polynomial stability?
- How to use Routh table to determine the stability?
- What’s Kharitonov Theorem?
- How to determine the stability of a closed-loop system?
- How to place the pole?

Appendix

sup v.s. max

A maximum is the largest number WITHIN a set.

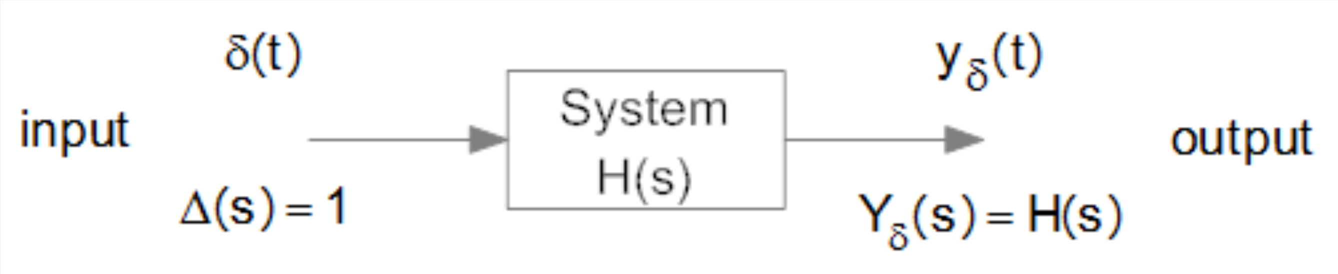


A sup is a number that BOUNDS a set.

A sup may or may not be part of the set itself (ZERO is not part of the set of negative numbers, but it is a sup because it is the least upper bound). If the sup IS part of the set, it is also the max.

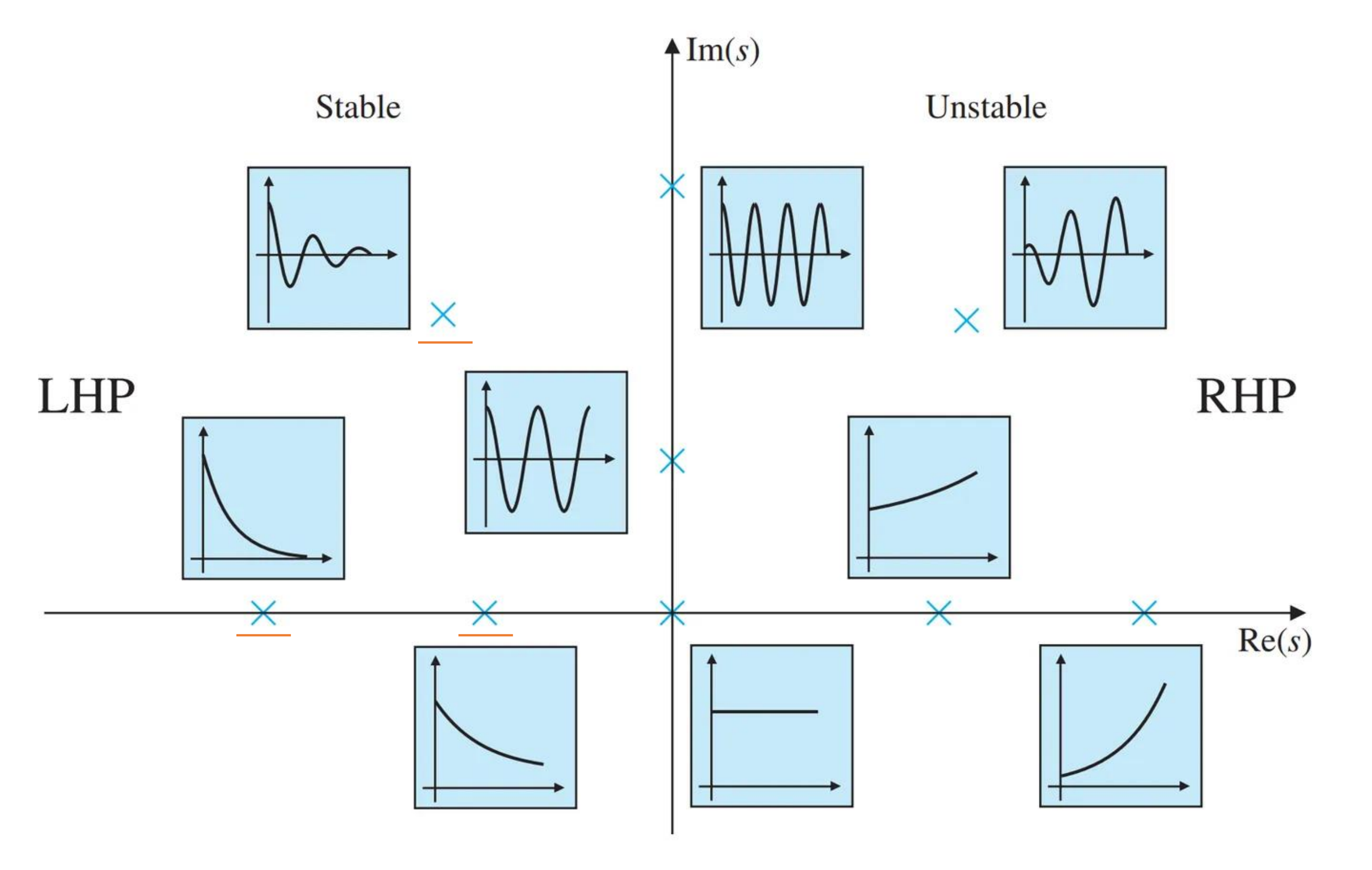
Impulse response function

Key Concept: The impulse response of a system is given by the transfer function. If the transfer function of a system is given by H(s), then the impulse response of a system is given by h(t) where h(t) is the inverse Laplace Transform of H(s).



$$h(t) = \mathcal{L}^{-1}\{H(s)\}$$

Relationship between pole location and transient response



More references: https://web.njit.edu/~mad29/refs/2ndorder_adfklhfw21.pdf

https://blog.csdn.net/zhuoqingjoking97298/article/details/125224676 (in Chinese)

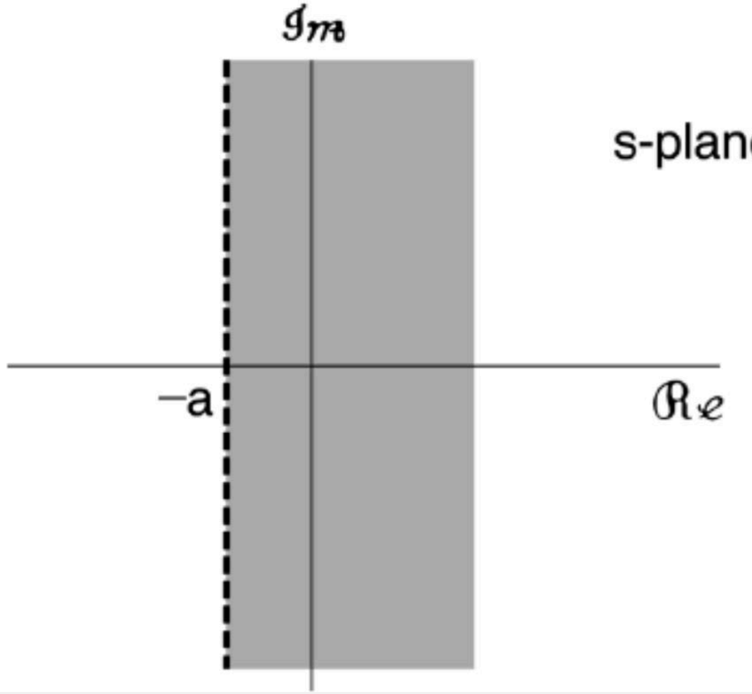
Relationship between pole location and Region of Convergence (ROC)

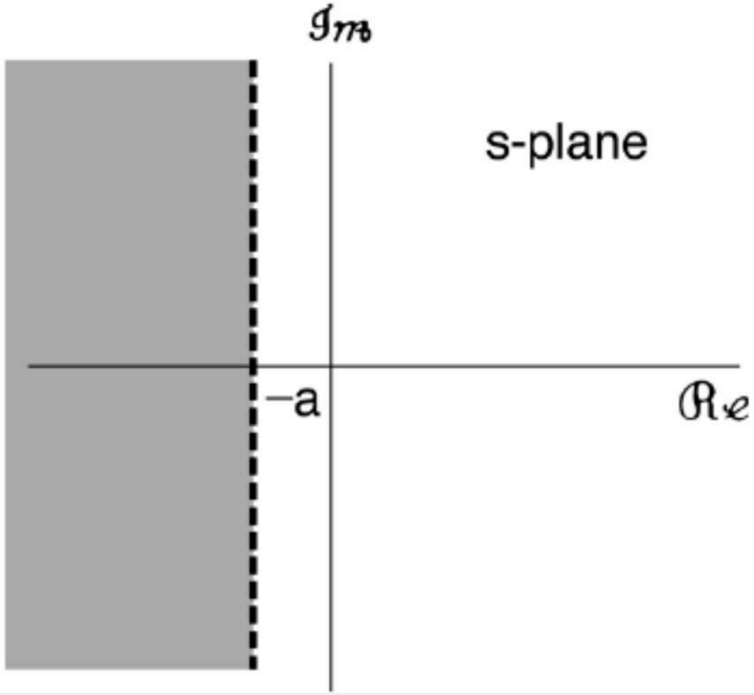
$X_1(s) = \dfrac{1}{s+a}, \quad \Re e\{s\} > -\Re e\{a\}$

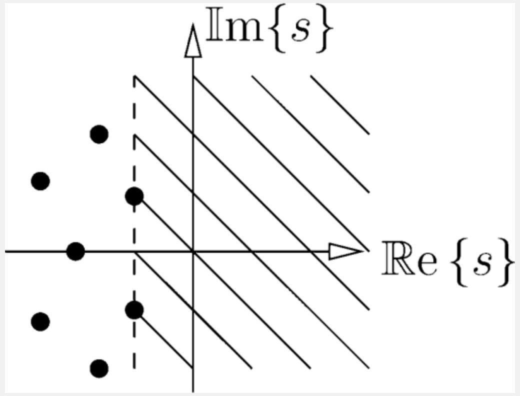
$x_1(t) = e^{-at}u(t)$ - right-sided signal

$X_2(s) = \dfrac{1}{s+a}, \quad \Re e\{s\} < -\Re e\{a\}$

$x_2(t) = -e^{-at}u(-t)$ - left-sided signal







Multiple poles

Note: in control theory, we only discuss the right-sided signal

More reference: https://www.youtube.com/watch?v=xOUVMngCGIs